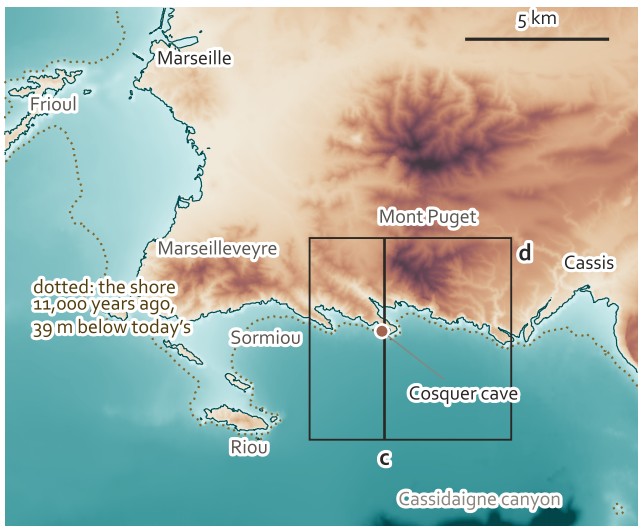
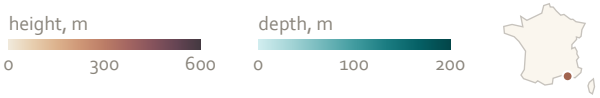
The sea rose 39 m in 11,000 years, 31 m of it in the first 3,500, and turned the valleys of the Calanques into inlets

Marseille to Cassis · relative sea level from 66 index points on the French Mediterranean coast (Vacchi et al. 2021) · ground from IGN RGE ALTI, sea floor from EMODnet

a The coast from Marseille to Cassis

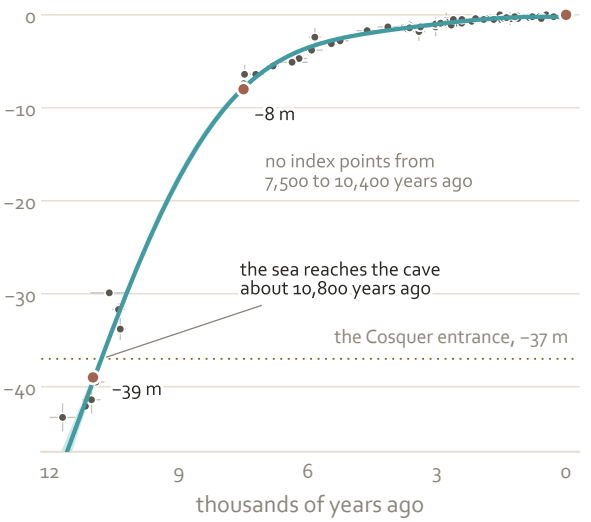
ground height and sea depth





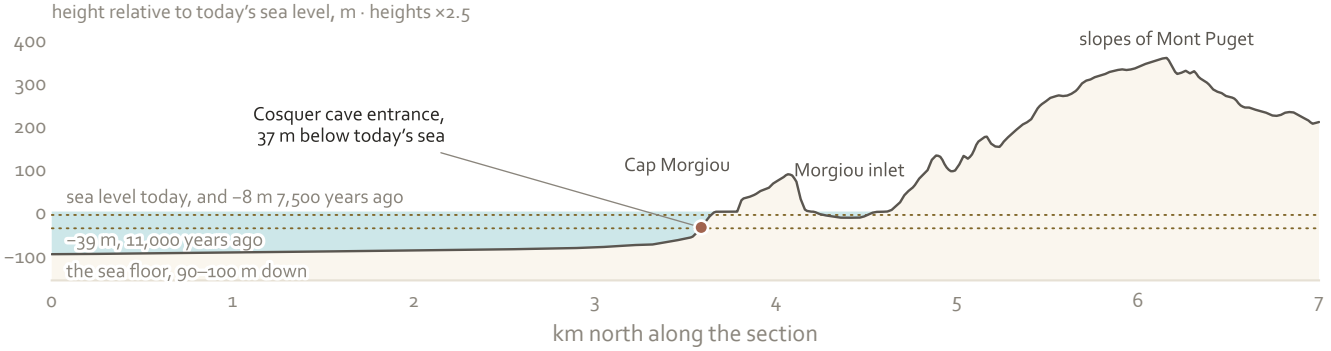
b The rise slowed from 89 cm a century to 11 cm

relative sea level, m



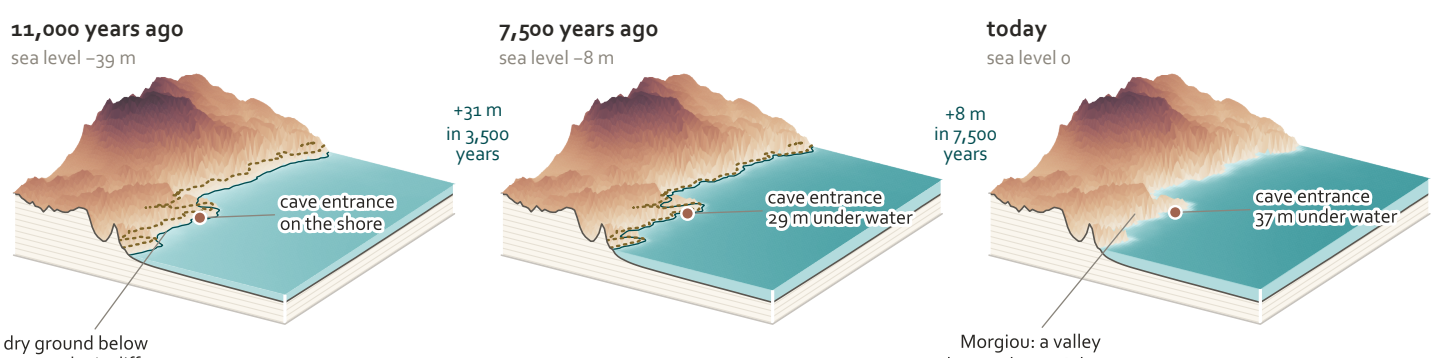
c Through Cap Morgiou: cliffs plunge to a floor that was dry land

height relative to today's sea level, m · heights ×2.5



d The sea climbed the cliffs, then filled the valleys

today's ground at three sea levels · heights ×3.5 · front face: the Sormiou valley; right face: the sea floor · dotted: today's shore



Data: IGN RGE ALTI (Licence Ouverte 2.0); EMODnet Bathymetry DTM (CC BY 4.0); sea-level index points, Vacchi et al. (2021), Zenodo (CC BY 4.0); Natural Earth. Lambert-93 · calanques.py · git bd6550c